

MySQL - Assignment – II

```
C1-89566-ABHISHEK [classworkdb]>select * from books;
```

id	name	author	subject	price
1001	Exploring C	Yashwant Kanetkar	C Programming	123.456
1002	Pointers in C	Yashwant Kanetkar	C Programming	371.019
1003	ANSI C Programming	E Balaguruswami	C Programming	334.215
1004	ANSI C Programming	Dennis Ritchie	C Programming	140.121
2001	C++ Complete Reference	Herbert Schildt	C++ Programming	417.764
2002	C++ Primer	Stanley Lippman	C++ Programming	620.665
2003	C++ Programming Language	Bjarne Stroustrup	C++ Programming	987.213
3001	Java Complete Reference	Herbert Schildt	Java Programming	525.121
3002	Core Java Volume I	Cay Horstmann	Java Programming	575.651
3003	Java Programming Language	James Gosling	Java Programming	458.238
4001	Operating System Concepts	Peter Galvin	Operating Systems	567.391
4002	Design of UNIX Operating System	Mauris J Bach	Operating Systems	421.938
4003	UNIX Internals	Uresh Vahalia	Operating Systems	352.822

```
13 rows in set (0.00 sec)
```

Q1. Display the total count of books in each subject using GROUP BY.

```
C1-89566-ABHISHEK [classworkdb]>select subject COUNT(*) AS Number_subject from books GROUP BY subject;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version 5.7.26 at line 1
```

```
C1-89566-ABHISHEK [classworkdb]>select subject, COUNT(*) AS Number_subject from books GROUP BY subject;
```

subject	Number_subject
C Programming	4
C++ Programming	3
Java Programming	3
Operating Systems	3

```
4 rows in set (0.00 sec)
```

Q2. Display the subjects that have more than 2 books using GROUP BY and HAVING.

```
C1-89566-ABHISHEK [classworkdb]>select subject, COUNT(*) AS Number_subject from books GROUP BY subject HAVING Number_subject > 2;
```

subject	Number_subject
C Programming	4
C++ Programming	3
Java Programming	3
Operating Systems	3

4 rows in set (0.00 sec)

Q3. Display the books in descending order of their price.

```
C1-89566-ABHISHEK [classworkdb]>select * from books order by price desc;
```

id	name	author	subject	price
2003	C++ Programming Language	Bjarne Stroustrup	C++ Programming	987.213
2002	C++ Primer	Stanley Lippman	C++ Programming	620.665
3002	Core Java Volume I	Cay Horstmann	Java Programming	575.651
4001	Operating System Concepts	Peter Galvin	Operating Systems	567.391
3001	Java Complete Reference	Herbert Schildt	Java Programming	525.121
3003	Java Programming Language	James Gosling	Java Programming	458.238
4002	Design of UNIX Operating System	Mauris J Bach	Operating Systems	421.938
2001	C++ Complete Reference	Herbert Schildt	C++ Programming	417.764
1002	Pointers in C	Yashwant Kanetkar	C Programming	371.019
4003	UNIX Internals	Uresh Vahalia	Operating Systems	352.822
1003	ANSI C Programming	E Balaguruswami	C Programming	334.215
1004	ANSI C Programming	Dennis Ritchie	C Programming	140.121
1001	Exploring C	Yashwant Kanetkar	C Programming	123.456

13 rows in set (0.00 sec)

Q4. Display the total number of books written by each author using GROUP BY.

```
C1-89566-ABHISHEK [classworkdb]>select author, COUNT(*) AS total_books from books GROUP BY author;
```

author	total_books
Yashwant Kanetkar	2
E Balaguruswami	1
Dennis Ritchie	1
Herbert Schildt	2
Stanley Lippman	1
Bjarne Stroustrup	1
Cay Horstmann	1
James Gosling	1
Peter Galvin	1
Mauris J Bach	1
Uresh Vahalia	1

```
11 rows in set (0.00 sec)
```

```
C1-89566-ABHISHEK [classworkdb]>
```

Q5. Display the authors who have written more than one book using GROUP BY and HAVING.

```
C1-89566-ABHISHEK [classworkdb]>select author, COUNT(*) AS total_books from books GROUP BY author HAVING total_books > 1;
```

author	total_books
Yashwant Kanetkar	2
Herbert Schildt	2

```
2 rows in set (0.00 sec)
```

```
C1-89566-ABHISHEK [classworkdb]>■
```


Q6. Display the subject-wise highest book price using GROUP BY.

```
C1-89566-ABHISHEK [classworkdb]>select * price, max(price) AS high_price from books GROUP BY price;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL
e' at line 1
C1-89566-ABHISHEK [classworkdb]>select subject, max(price) AS high_price from books GROUP BY subject as
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL
C1-89566-ABHISHEK [classworkdb]>select subject, max(price) AS high_price from books GROUP BY subject;

+-----+-----+
| subject          | high_price |
+-----+-----+
| C Programming    | 371.019    |
| C++ Programming  | 987.213    |
| Java Programming | 575.651    |
| Operating Systems | 567.391    |
+-----+-----+
4 rows in set (0.00 sec)

C1-89566-ABHISHEK [classworkdb]>.
```

Q7. Display the subject-wise average book price using GROUP BY.

```
C1-89566-ABHISHEK [classworkdb]>select subject, AVG(price) AS avg_price from books GROUP BY subject;

+-----+-----+
| subject          | avg_price  |
+-----+-----+
| C Programming    | 242.2027531 |
| C++ Programming  | 675.2139994 |
| Java Programming | 519.6699931 |
| Operating Systems | 447.3836568 |
+-----+-----+
4 rows in set (0.00 sec)

C1-89566-ABHISHEK [classworkdb]>
```

Q8. Display the books where the price is higher than the average price of their respective subject using GROUP BY and HAVING.

```
C1-89566-ABHISHEK [classworkdb]>select price, avg(price) from books GROUP BY subject HAVING avg(price) < price;
ERROR 1055 (42000): Expression #1 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'classworkdb.books.price' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by
C1-89566-ABHISHEK [classworkdb]>select avg(price) from books;
+-----+
| avg(price) |
+-----+
| 453.5087662 |
+-----+
1 row in set (0.00 sec)

C1-89566-ABHISHEK [classworkdb]>select price, avg(price) from books GROUP BY subject HAVING price > 453;
ERROR 1055 (42000): Expression #1 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'classworkdb.books.price' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by
C1-89566-ABHISHEK [classworkdb]>select subject, price, avg(price) from books GROUP BY subject HAVING price > 453;
ERROR 1055 (42000): Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'classworkdb.books.price' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by
C1-89566-ABHISHEK [classworkdb]>select subject, max(price) AS highest_price, AVG(price) AS average_price from books GROUP BY subject HAVING AVG(price) > 453;
+-----+-----+-----+
| subject | highest_price | average_price |
+-----+-----+-----+
| C++ Programming | 987.213 | 675.2139994 |
| Java Programming | 575.651 | 519.6699931 |
+-----+-----+-----+
2 rows in set (0.00 sec)

C1-89566-ABHISHEK [classworkdb]>.
```

Q9. Display the books in ascending order of the author's name.

```
C1-89566-ABHISHEK [classworkdb]>select * from books ORDER BY author asc;
+-----+-----+-----+-----+-----+
| id | name | author | subject | price |
+-----+-----+-----+-----+-----+
| 2003 | C++ Programming Language | Bjarne Stroustrup | C++ Programming | 987.213 |
| 3002 | Core Java Volume I | Cay Horstmann | Java Programming | 575.651 |
| 1004 | ANSI C Programming | Dennis Ritchie | C Programming | 140.121 |
| 1003 | ANSI C Programming | E Balaguruswami | C Programming | 334.215 |
| 2001 | C++ Complete Reference | Herbert Schildt | C++ Programming | 417.764 |
| 3001 | Java Complete Reference | Herbert Schildt | Java Programming | 525.121 |
| 3003 | Java Programming Language | James Gosling | Java Programming | 458.238 |
| 4002 | Design of UNIX Operating System | Mauris J Bach | Operating Systems | 421.938 |
| 4001 | Operating System Concepts | Peter Galvin | Operating Systems | 567.391 |
| 2002 | C++ Primer | Stanley Lippman | C++ Programming | 620.665 |
| 4003 | UNIX Internals | Uresh Vahalia | Operating Systems | 352.822 |
| 1001 | Exploring C | Yashwant Kanetkar | C Programming | 123.456 |
| 1002 | Pointers in C | Yashwant Kanetkar | C Programming | 371.019 |
+-----+-----+-----+-----+-----+
13 rows in set (0.00 sec)

C1-89566-ABHISHEK [classworkdb]>
```

Q10. Display the subject-wise sum of book prices and filter the subjects where the sum of prices is more than 1000 using GROUP BY and HAVING.

```
C1-89566-ABHISHEK [classworkdb]>select subject, SUM(price) AS total_price
-> from books
-> group by subject
-> HAVING SUM(price) > 1000;
+-----+-----+
| subject          | total_price |
+-----+-----+
| C++ Programming  | 2025.642    |
| Java Programming | 1559.010    |
| Operating Systems| 1342.151    |
+-----+-----+
3 rows in set (0.00 sec)

C1-89566-ABHISHEK [classworkdb]>
```